

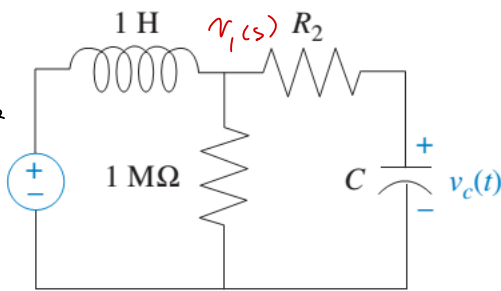
Problem 1:

For the circuit shown in Figure, find the values of R_2 and C to yield 8% overshoot with a settling time of 1 ms for the voltage across the capacitor, with $v_i(t)$ as a step input.

need to find

$$\frac{V_c(s)}{V_i(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Where $V_c(s) = \frac{1}{s} v_i(t)$



FIGURE

The given settling time is a 2% settling time.

$$\%OS = \frac{y_{max} - y_{ss}}{y_{ss}} \times 100\%$$

$$t_s (2\%) = 1 \text{ ms} = 4\tau$$

$$\Rightarrow \tau = 0.25 \text{ ms}$$

$$\%OS = 100e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$$

$$\zeta = \left| \ln \left(\frac{\%OS}{100} \right) \right| / \sqrt{\pi^2 + \left(\ln \left(\frac{\%OS}{100} \right) \right)^2}$$

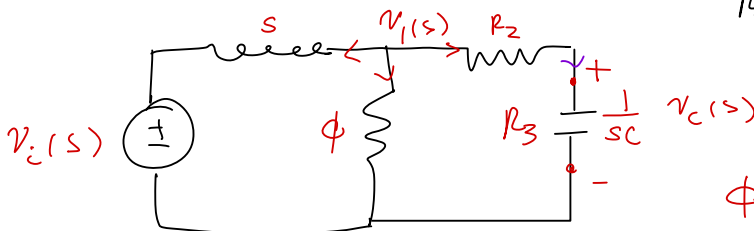
$$\zeta = 0.6266$$

$$\tau = \frac{1}{\zeta\omega_n}$$

$$\omega_n = \frac{1}{\tau\zeta}$$

$$\omega_n = \frac{1}{(0.25 \text{ ms})(0.6266)}$$

$$\omega_n = 6386.6578 \text{ rad/s}$$



$$R_{eq} = R_2 + \frac{1}{sC}$$

$$\phi = 1 \mu\Omega$$

$$(1) \frac{V_1(s) - V_c(s)}{R_2} = sC V_c(s)$$

$$V_1(s) - V_c(s) = sC R_2 V_c(s)$$

$$V_c(s) C (1 + sC R_2) = V_1(s) -$$

$$(2) \left(\frac{V_1(s) - V_c(s)}{s} + \frac{V_1(s)}{\phi} + \frac{V_1(s) - V_c(s)}{R_2} \right) s\phi R_2$$

$$(V_1(s) - V_c(s))(\phi R_2) + V_1(s)(sR_2) + (V_1(s) - V_c(s))(s\phi) = 0$$

$$\cancel{V_1(s)\phi R_2} - \cancel{V_c(s)\phi R_2} + \cancel{V_1(s)sR_2} + \cancel{V_1(s)s\phi} - \cancel{V_c(s)s\phi} = 0$$

$$V_1(s)(\phi R_2 + sR_2 + s\phi) - V_c(s)\phi R_2 - V_c(s)s\phi = 0$$

$$V_i(s) = V_c(s)(1 + sCR_2)$$

$$V_c(s)(1 + sCR_2)(\phi R_2 + sR_2 + s\phi) - V_i(s)\phi R_2 - V_c(s)s\phi = 0$$

$$V_c(s)(\cancel{\phi R_2 + sR_2 + s\phi} + \cancel{sCR_2^2\phi} + \cancel{s^2CR_2^2} + \cancel{s^2\phi R_2\phi} - \cancel{s\phi}) = V_i(s)\phi R_2$$

$$V_c(s)(s^2(CR_2^2 + CR_2\phi) + s(R_2 + CR_2^2\phi) + \phi R_2) = V_i(s)\phi R_2$$

$$\begin{aligned} \frac{V_c(s)}{V_i(s)} &= \frac{\phi R_2}{s^2(CR_2^2 + CR_2\phi) + s(R_2 + CR_2^2\phi) + \phi R_2} \\ &= \frac{\phi}{s^2(CR_2 + C\phi) + s(1 + CR_2\phi) + \phi} \\ &= \frac{\phi / CR_2 + C\phi}{s^2 + s\left(\frac{1 + CR_2\phi}{CR_2 + C\phi}\right) + \frac{\phi}{CR_2 + C\phi}} \end{aligned}$$

$$(1) \omega_n^2 = \frac{\phi}{CR_2 + C\phi} \Rightarrow C = \frac{\phi}{\omega_n^2(R_2 + \phi)}$$

$$(2) 2\zeta\omega_n = \frac{1 + CR_2\phi}{CR_2 + C\phi} \quad T = \frac{1}{\zeta\omega_n}$$

$$\zeta\omega_n C(R_2 + \phi) = 1 + CR_2\phi$$

$$1, 2\zeta\omega_n CR_2 + 2\zeta\omega_n C\phi - CR_2\phi = 1$$

$$C(2\zeta\omega_n R_2 + 2\zeta\omega_n \phi - R_2\phi) = 1$$

$$\frac{\phi}{\omega_n^2(R_2 + \phi)}(2\zeta\omega_n R_2 + 2\zeta\omega_n \phi - R_2\phi) = 1$$

$$2\cancel{\phi\zeta\omega_n R_2 - R_2\phi^2} + 2\zeta\omega_n\phi^2 = \cancel{\omega_n^2 R_2} + \omega_n^2\phi$$

$$R_2(2\phi\zeta\omega_n - \phi^2 - \omega_n^2) = \omega_n^2\phi - 2\zeta\omega_n\phi^2$$

$$R_2 = \frac{\omega_n^2\phi - 2\zeta\omega_n\phi^2}{2\phi\zeta\omega_n - \phi^2 - \omega_n^2}$$

$$R_2 = 8023.0679 \Omega$$

$$C = 24.3210 \text{ nF}$$